



COURSE SYLLABUS

Course Number:	CSB280H1F	Term:	Fall 2026
Course Title:	Data Science for Cell and Systems Biology	Office Hours and location:	Post-class, OI-2214
Course Day/Time:	Thursdays, 1500h-1700h	Location:	OI-2214
Instructor:	Calvin Mok	Email:	calvin.mok@utoronto.ca
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Teaching Assistant:	Cameron Parro	Email:	c.parro@mail.utoronto.ca
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Course Administrator:	If applicable	Email:	If applicable

Communication and Quercus

This Course uses the University's learning management system, Quercus, to post information about the course. This includes posting readings and other materials required to complete class activities and course assignments, as well as sharing important announcements and updates. New information and resources will be posted regularly as we move through the term. To access the course website, go to the U of T Quercus log-in page at <https://q.utoronto.ca>.

The course will also use the resources available at <https://datatools.utoronto.ca> to work through lectures and tutorial materials. A specific link for synchronizing course data will be provided in Quercus.

Instructors will be available for a short period after lectures to discuss questions regarding class material. Any administrative questions (e.g. tutorials, missed quizzes, accommodation requests) can be directed to the instructors or teaching assistants via email.

SPECIAL NOTE ABOUT GRADES POSTED ONLINE: Please also note that any grades posted are for your information only, so you can view and track your progress through the course. No grades are considered official, including any posted in Quercus at any point in the term, until they have been formally approved and posted on ACORN at the end of the course. Please contact me as soon as possible if you think there is an error in any grade posted on Quercus.

Course Requirements

Prerequisites: None

Exclusion: STA272H1, STA3012H1

Recommended Preparation: EEB125H1 or GGR274H1



Course Description

The abundance of data in biological sciences continues to grow year after year. The skills required to navigate and thrive in this field are no longer confined to the laboratory bench as experimental results go beyond simple analyses. This course will teach introductory programming skills, and the conceptual tools used in the analysis of big data such as dimensional reduction, visualization, and machine learning. Students will get practical experience writing code to analyze example datasets.

Students will require access to a laptop computer. No programming experience is required.

Learning Outcomes

This is a beginner's introduction to data science using the programming language R. It is aimed at individuals with some background in biology but no prior experience in data science. Individuals who complete the course will be able to:

- Work with the RStudio and Markdown Notebook environment and navigate the R programming language.
- Understand data structures and data types needed for systems biology.
- Import data into R and manipulate tabular data.
- Learn how to present biological data using graphs and plots.
- Work with high-dimensional biological datasets to identify trends and groupings.
- Formulate simple quantitative models and test hypotheses by fitting them to data.

Required Course Material

Throughout the course we'll work with sets of data that take us through the various steps of analysis from importing to data wrangling (formatting) to visualization. Each class will consist of a short introductory section followed by 'code-along' hands-on learning that will gradually build up the lecture's topic(s). Students are expected to have access to a computer during class and are encouraged to ask questions while coding along with the instructor.

Tutorials will consist of materials for practicing, reinforcing, or expanding on lecture information after class each week. There will also be 4 quiz/assessments taking place during these tutorials with their content covering specific portions of the course.

Course material references concepts from:

- "R for Data Science" by Hadley Wickham and Garrett Grolemund (<http://r4ds.had.co.nz/>)
- Statistical Modeling and Machine Learning for Molecular Biology by Alan M. Moses

Relevant Dates

Please visit the Arts and Science website for a list of academic dates and deadlines:

<https://www.artsci.utoronto.ca/current/dates-deadlines/academic-dates>



Topics

Part 1: Introduction to R

Lecturer: Calvin Mok

Week	Lecture Topic	Subtopics
1	Introduction to where data comes from and how to handle it	• Where data comes from • Best practice for coding • Introducing the basics of R
2	Storing information in data structures	• Basic data types • Basic data structures • Factors and categorical variables
3	Importing data and working with data frames	• Importing tabular data • Reformatting data with dplyr
4	Working with tidy data	• Rules for long-format/tidy data • Exploring and summarizing data
5	Visualizing data	• The grammar of graphics with ggplot2 • How to make and interpret common plots
6	Dimensional reduction of big data	• Working with large datasets • Finding patterns using clustering • Dimensional reduction techniques • Non-linear projection of large datasets

Part 2: Models, predictions, and classifiers

Lecturer: Alan M. Moses

Week	Lecture Topic	Subtopics
7	Models	• Parameters, estimation, objective functions • Predicting protein levels from mRNA • Predicting phenotype from genotype
8	High-dimensional models and sparse multiple regression	• Prediction of phenotype using many genotypes • Holding out data • Prediction of cell type using RNA-seq
9	Probability and correlation	• Noise in data, measures of confidence/uncertainty • Regression as a probabilistic model • Correlation and hypothesis testing
10	Probability in high dimensions	• Multiple testing • Multivariate distributions and correlated data
11	Classification	• Going beyond R-squared and normal distributions



		• Confusion matrix • Measures of predictive power
12	All models are wrong. Is mine useful?	• Holding out data carefully: data splits in Cell & Systems Biology • Data balance and 'use-case' • Prediction vs. understanding in high dimensions • AlphaFold

Evaluation

Item	Note	% Mark	Due Date
Completed R Markdown Lecture Notebook	12 lectures x 2% each	24%	2359h each Sunday following lecture
Tutorial Quiz*	4 assessments x 10% each Closed-book, hand-written	40%	Quiz 1: Tutorial 3, September 29/30 Quiz 2: Tutorial 5, October 13/14 Quiz 3: Tutorial 8, November 10/11 Quiz 4: Tutorial 10, December 1/2
Final exam	Covers all sections of the course	36%	TBA

* Note that quizzes will be somewhat cumulative, although each will aim to focus on more recent lectures.

Assignment submission policy

Completed lecture skeletons should be submitted in their original R markdown format (RMD) through the provided assignment portals on Quercus. Alternative formats will not be accepted.

Late assignment penalty policy

Late submissions of R markdown notebooks, will be given a mark of 0. Accommodations may be made under extenuating conditions, usually as recommended by the college registrar after discussion of acceptable supporting documentation.

Missed test/quiz policy

Students that have missed a test/quiz must go to their college registrar to discuss their absence and present any required documentation. If the college registrar recommends accommodation (via email to instructors), then tutorial quiz weights will be adjusted appropriately. This



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adjustment will only apply to the first missed quiz. Without appropriate communication from the registrar any missed quizzes will be given a mark of 0. Any missed quizzes past the first missed quiz will be given a mark of 0.

Lab/Tutorial Information

Students are expected to attend their assigned tutorial sections. Tutorials will include quizzes as specified in the evaluation section. If students know ahead of time that they will be unable to attend their tutorial/quiz section (for legitimate reasons), they must coordinate with their section's teaching assistant **and** the alternate section's teaching assistant to obtain approval to temporarily switch.

Mental Health and Well-Being

Your health and well-being in this course are important to everyone – from the instructors, TAs, administrative staff and the University of Toronto as a whole. The following are suggestions and resources to make sure you have a productive and healthy semester. The university experience can be a challenging one, there is no need to go about it alone. If you or anyone you know could use someone to talk to (or text with), here are some resources in addition to your instructors, program coordinators, and TAs:

- Your College Registrar: uoft.me/registrars
- UofT Health and Wellness: 700 Bay Street Toronto, ON M5G 1Z6
416-978-8030 | info.hwc@utoronto.ca

24/7 emergency counselling services:

- [U of T Telus Health Student Support](#) (formerly My SSP) | 1-844-451-9700. Outside of North America, call 001-416-380-6578.
Culturally-competent mental health and counselling services in 146 languages for all U of T students.
- [Good2Talk Student Helpline](#) | 1-866-925-5454
Professional counseling, information and referrals helpline for mental health, addictions and students well-being.

Acorn Absence Declaration

During the academic term, if you are absent from your studies and unable to complete course work, you should follow the **missed test policy** outlined in this syllabus. Normally, I will ask you for documentation in support of your specific medical circumstances. This documentation can be an Absence Declaration (via ACORN) or the University's Verification of Student Illness or Injury (VOI) form. The VOI indicates the impact and severity of the illness, while protecting your privacy about the details of the nature of the illness. If you cannot submit a VOI due to limits on terms of use, you can submit a different form (like a letter from a doctor), as long as it is an original document, and it contains the same information as the VOI (including dates, academic impact, practitioner's signature, phone and registration number). For more information on the VOI,



please see <https://registrar.utoronto.ca/policies-and-guidelines/verification-of-illness-or-injury/>. For information on Absence Declaration Tool for A&S students, please see <https://www.artsci.utoronto.ca/absence>. If you get a concussion, break your hand, or suffer some other acute injury, you should register with Accessibility Services as soon as possible. For extended absences and for absences due to non-medical reasons, make sure to contact your [College Registrar's Office](#).

Use of a Plagiarism Detection Tool for Course Assignments

Normally, students will be required to submit their course essays to the University's plagiarism detection tool for a review of textual similarity and detection of possible plagiarism. In doing so, students will allow their essays to be included as source documents in the tool's reference database, where they will be used solely for the purpose of detecting plagiarism. The terms that apply to the University's use of this tool are described on the Centre for Teaching Support & Innovation web site. (<https://uoft.me/pdt-faq>).

Academic Integrity

Academic integrity is a foundation of our university community's intellectual life. What does it mean to act with academic integrity? Acting with academic integrity means not 'cheating' to get ahead. U of T supports the International Center for Academic Integrity's definition of academic integrity as acting with honesty, trust, fairness, respect, responsibility, and courage in all academic matters. <https://www.academicintegrity.utoronto.ca/>

The University of Toronto treats cases of academic misconduct very seriously. Academic integrity is a fundamental value of learning and scholarship at the U of T. Participating honestly, respectfully, responsibly, and fairly in this academic community ensures that your U of T degree is valued and respected as a true signifier of your individual academic achievement.

The University of Toronto's [Code of Behaviour on Academic Matters](#) outlines the behaviours that constitute academic misconduct, the processes for addressing academic offences, and the penalties that may be imposed. You are expected to be familiar with the contents of this document. Potential offences include, but are not limited to:

In papers and assignments:

- using someone else's ideas or words without appropriate acknowledgement
- submitting your own work in more than one course without the permission of the instructor
- making up sources or facts
- obtaining or providing unauthorized assistance on any assignment (this includes working in groups on assignments that are supposed to be individual work)

On tests and exams:

- using or possessing any unauthorized aid, including a cell phone
- looking at someone else's answers



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- letting someone else look at your answers
- misrepresenting your identity
- submitting an altered test for re-grading

Misrepresentation:

- falsifying or altering any documentation required by the University, including (but not limited to) doctor's notes
- falsifying institutional documents or grades

Avoiding Plagiarism

<https://advice.writing.utoronto.ca/using-sources/how-not-to-plagiarize/>

Generative AI:

It is not required to complete any coursework, and we caution you to not rely on those tools to complete your coursework. Instead, you can treat generative AI as a supplementary tool for exploration and for gathering information from across sources and assimilating for understanding. The final submitted assignment must be original work produced by the individual student alone. Ultimately, you (and not any AI tool) are responsible for your own learning in this course, and for all the work you submit for credit. It is your responsibility to critically evaluate the content generated, and to regularly assess your own learning independent of any generative AI tools. Over reliance on generative AI may give you a false sense of how much you have learned, which can lead to poor performance on assignments and exams, or in future work/study after graduation.

All suspected cases of academic misconduct will be investigated following the procedures outlined in the [Code of Behaviour on Academic Matters](#). If you have any questions about what is or is not permitted in this course, please do not hesitate to contact me. If you have questions about appropriate research and citation methods, you are expected to seek out additional information from me or other available campus resources like the [College Writing Centres](#), the [Academic Success Centre](#), or the [U of T Writing Website](#).

Accommodations

Students with Disabilities:

Students with diverse learning styles and needs are welcome in this course. If you have an acute or ongoing disability issue or accommodation need, you should register with Accessibility Services (AS) at the beginning of the academic year by visiting <https://studentlife.utoronto.ca/departments/accessibility-services/>. Without registration, you will not be able to verify your situation with your instructors, and instructors will not be advised about your accommodation needs. AS will assess your situation, develop an accommodation plan with you, and support you in requesting accommodation for your course work. Remember that the process of accommodation is private: AS will not share details of your needs or condition with any instructor, and your instructors will not reveal that you are registered with AS.



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Accessibility Services: accessibility.services@utoronto.ca or
<https://studentlife.utoronto.ca/departments/accessibility-services/>

Religious Accommodations:

As a student at the University of Toronto, you are part of a diverse community that welcomes and includes students and faculty from a wide range of cultural and religious traditions. For my part, I will make every reasonable effort to avoid scheduling tests, examinations, or other compulsory activities on religious holy days not captured by statutory holidays. Further to University Policy, if you anticipate being absent from class or missing a major course activity (such as a test or in-class assignment) due to a religious observance, please let me know as early in the course as possible, and with sufficient notice (at least two to three weeks), so that we can work together to make alternate arrangements.

Family Care Responsibilities:

The University of Toronto strives to provide a family-friendly environment. You may wish to inform me if you are a student with family responsibilities. If you are a student parent or have family responsibilities, you also may wish to visit the Family Care Office website at familycare.utoronto.ca.

Equity, Diversity and Inclusion

The University of Toronto is committed to equity, human rights and respect for diversity. All members of the learning environment in this course should strive to create an atmosphere of mutual respect where all members of our community can express themselves, engage with each other, and respect one another's differences. U of T does not condone discrimination or harassment against any persons or communities.